Solutions of Diophanite Equation

$x^3+y^3=x^2+kxy+y^2$

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Abstract

First time I saw problem of similar form while listing AOPS. The original statement of the problem was as follows:

BMO 2017/1 Find all ordered pairs of positive integers(x,y) such that:
$$x^3+y^3=x^2+42xy+y^2.$$

After that, We became interested in studying solutions to the generalized case. Thus we study integer solutions of equation $x^3+y^3=x^2+kxy+y^2$

Introduction

For small cases:

1. if $k=0$,

$$x^3+y^3=x^2+y^2$$

$$x^2-xy+y^2|x^2+y^2$$

Then

$$x^2-xy+y^2|xy$$

So $x = y = 0$ and $x = y = 1$ are only solutions

2. if $k = -1$, similarly we can get answers are all x, y such that $x + y = 1$

3. if $k = 1$,

$$x^3 + y^3 = x^2 + xy + y^2$$

Let

$$s = x + y, \quad p = xy.$$

Then

$$x^3 + y^3 = (x+y)^3 - 3xy(x+y) = s^3 - 3ps,$$

$$x^2 + xy + y^2 = (x+y)^2 - xy = s^2 - p.$$

Substituting, we get

$$s^3 - 3ps = s^2 - p.$$

Rearranging:

$$s^3 - s^2 - 3ps + p = 0,$$

$$s^2(s-1) - p(3s-1) = 0.$$

Thus,

$$p(3s-1) = s^2(s-1).$$

Observe that

$$\gcd(3s-1, s) = 1,$$

since any common divisor divides

$$(3s-1) - 3s = -1.$$

Also,

$$\gcd(3s-1, s-1) \mid ((3s-1) - 3(s-1)) = 2.$$

Hence from

$$3s-1 \mid s^2(s-1)$$

it follows that

$$3s-1 \mid 2.$$

Therefore,

$$3s-1 \in \{-2,-1,1,2\}.$$

Checking integer values:

$$3s-1=-1 \Rightarrow s=0, \quad 3s-1=2 \Rightarrow s=1.$$

Case 1: $s=0$

$$p(-1)=0 \Rightarrow p=0.$$

Thus,

$$x+y=0, \quad xy=0 \Rightarrow x=y=0.$$

Case 2: $s=1$

$$p(2)=0 \Rightarrow p=0.$$

Thus,

$$x+y=1, \quad xy=0,$$

so

$$(x,y)=(1,0),\ (0,1).$$

Answer:

$$(x,y)=(0,0),\ (1,0),\ (0,1).$$

4. if $k=2$, then the equation becomes

$$x^3+y^3=x^2+2xy+y^2=(x+y)^2,$$

and every pair with

$$x+y=0$$

is a solution, so there are infinitely many integer solutions.

Theorems and their proves.

Despite the small number of solutions for small cases, an equation of this type has quite a lot of solutions.

Theorem 1: For every positive integers n, l, there is exist $k > l$, such that

$$x^3 + y^3 = x^2 + kxy + y^2$$

has more than n solutions in $\mathbb{Z}$.

Proof. We look for solutions of the form

$$y = mx$$

for some positive integer m. Substituting into

$$x^3 + y^3 = x^2 + kxy + y^2$$

gives

$$x^3 + (mx)^3 = x^2 + kx(mx) + (mx)^2,$$

hence

$$x^3(1 + m^3) = x^2(1 + km + m^2).$$

Therefore, if

$$1 + m^3 \mid 1 + km + m^2,$$

then

$$x = \frac{1 + km + m^2}{1 + m^3}, \qquad y = mx$$

is an integer solution.

So for each chosen m, it is enough to require

$$1 + km + m^2 \equiv 0 \pmod{1 + m^3}.$$

A sufficient condition is

$$k \equiv m(m-1) \pmod{1+m^3},$$

because then

$$1+km+m^2 \equiv 1+m^2(m-1)+m^2 = 1+m^3 \equiv 0 \pmod{1+m^3}.$$

Now choose

$$m_i = 2^{2^i} \qquad (i = 1,2,\ldots,n+1),$$

and let

$$q_i = 1+m_i^3 = 1+2^{3\cdot 2^i}.$$

We claim that the integers $q_1, q_2, \ldots, q_{n+1}$ are pairwise coprime. Indeed, suppose $i < j$ and some integer $d > 1$ divides both q_i and q_j. Then

$$2^{3\cdot 2^i} \equiv -1 \pmod{d}.$$

Raising both sides to the power 2^{j-i}, we obtain

$$2^{3\cdot 2^j} \equiv (-1)^{2^{j-i}} = 1 \pmod{d}.$$

On the other hand, since $d \mid q_j$, we also have

$$2^{3\cdot 2^j} \equiv -1 \pmod{d}.$$

This is impossible. Hence

$$\gcd(q_i, q_j) = 1 \qquad \text{for all } i \neq j.$$

Therefore, by the Chinese Remainder Theorem, there exists an integer k such that

$$k \equiv m_i(m_i-1) \pmod{q_i} \qquad (i = 1,2,\ldots,n+1).$$

Moreover, all such integers form an arithmetic progression, so we may choose one with

$$k > l.$$

For each i, define

$$x_i = \frac{1 + km_i + m_i^2}{1 + m_i^3}, \qquad y_i = m_i x_i.$$

By construction, $x_i \in \mathbb{Z}$, hence $(x_i, y_i) \in \mathbb{Z}^2$. Also,

$$x_i^3 + y_i^3 = x_i^3(1 + m_i^3) = x_i^2(1 + km_i + m_i^2) = x_i^2 + kx_i y_i + y_i^2,$$

so (x_i, y_i) is a solution of the equation.

These solutions are distinct, since

$$\frac{y_i}{x_i} = m_i$$

and the numbers $m_1, m_2, \ldots, m_{n+1}$ are pairwise distinct.

Thus the equation has at least $n+1$ distinct integer solutions. In particular, it has more than n solutions in $\mathbb{Z}$.

This proves the Theorem 1.

Theorem 2: Let

$$N_k = \#\{(x, y) \in \mathbb{Z}^2 : \ x^3 + y^3 = x^2 + kxy + y^2\}.$$

If $k \neq -1, 2$, then

$$N_k \leq 4\,\tau\Big(\Big|(k-2)^2(k+1)\Big|\Big),$$

where $\tau(m)$ denotes the number of positive divisors of m.

Moreover, for $k = -1$ and $k = 2$, the equation has infinitely many integer solutions.

Proof. Consider the equation

$$x^3 + y^3 = x^2 + kxy + y^2.$$

Set

$$s = x + y, \qquad p = xy.$$

Then

$$x^3 + y^3 = (x+y)^3 - 3xy(x+y) = s^3 - 3ps,$$

and

$$x^2 + kxy + y^2 = (x+y)^2 + (k-2)xy = s^2 + (k-2)p.$$

Hence

$$s^3 - 3ps = s^2 + (k-2)p,$$

so

$$s^3 - s^2 = p(3s + k - 2),$$

that is,

$$s^2(s-1) = p(3s + k - 2).$$

Let

$$D = 3s + k - 2.$$

Then

$$D \mid s^2(s-1).$$

We first note that if $D = 0$, then

$$s^2(s-1) = 0,$$

so $s = 0$ or $s = 1$. Since $D = 3s + k - 2$, this gives $k = 2$ or $k = -1$. Therefore, for

$$k \neq -1, 2,$$

we must have

$$D \neq 0.$$

We now prove that

$$|D| \mid |(k-2)^2(k+1)|.$$

Let $q^a \parallel D$, where q is a prime. Since

$$D \mid s^2(s-1)$$

and

$$\gcd(s, s-1) = 1,$$

the prime q divides exactly one of s and $s-1$.

Case 1: $q \mid s-1$.

Since $q \nmid s$, from $q^a \mid s^2(s-1)$ we get

$$q^a \mid (s-1).$$

But also $q^a \mid D$, hence

$$q^a \mid D - 3(s-1) = k+1.$$

Case 2: $q \mid s$.

Write

$$q^b \parallel s.$$

Since $q \nmid s-1$, from $q^a \mid s^2(s-1)$ we obtain

$$a \leq 2b.$$

Also, because $q^a \mid D$ and $q^b \mid s$, we have

$$q^{\min(a,b)} \mid \gcd(D, s).$$

But

$$D - 3s = k-2,$$

so

$$q^{\min(a,b)} \mid (k-2).$$

If $a \leq b$, then $q^a \mid (k-2)$. If $a > b$, then $q^b \mid (k-2)$, and since $a \leq 2b$, it follows that

$$q^a \mid (k-2)^2.$$

Thus in all cases,

$$q^a \mid (k-2)^2.$$

Therefore, every prime power divisor of D divides either $k+1$ or $(k-2)^2$, and so

$$|D| \mid |(k-2)^2(k+1)|.$$

Now let

$$M = (k-2)^2(k+1).$$

Every integer solution (x,y) gives rise to a divisor D of M, and $s = x+y$ is then uniquely determined by

$$s = \frac{D-k+2}{3}.$$

After that, $p = xy$ is uniquely determined from

$$p = \frac{s^2(s-1)}{D}.$$

Finally, x and y are the roots of

$$t^2 - st + p = 0.$$

For fixed s and p, this quadratic gives at most two ordered integer pairs (x,y), namely (x,y) and (y,x).

Since $M \neq 0$ for $k \neq -1, 2$, the number of integer divisors of M is

$$2\tau(|M|).$$

Hence

$$N_k \leq 2 \cdot 2\tau(|M|) = 4\tau(|M|).$$

Therefore

$$N_k \leq 4\,\tau\Big(\Big|(k-2)^2(k+1)\Big|\Big).$$

It remains to note the exceptional cases from introduction. This completes the proof.

Numericial examples

Below, all solution sets are sets of ordered pairs $(x,y) \in \mathbb{Z}^2$.

k	Integer solutions of $x^3+y^3=x^2+kxy+y^2$
-5	$(0,0),\ (1,0),\ (0,1)$
-4	$(0,0),\ (1,0),\ (0,1),\ (-1,-1)$
-1	infinitely many: $(x,y)=(t,1-t)$, $t \in \mathbb{Z}$
0	$(0,0),\ (1,0),\ (0,1),\ (1,1)$
1	$(0,0),\ (1,0),\ (0,1)$
2	infinitely many: $(x,y)=(t,-t)$, $t \in \mathbb{Z}$
4	$(0,0),\ (1,0),\ (0,1),\ (3,3)$
6	$(0,0),\ (1,0),\ (0,1),\ (-2,1),\ (1,-2),\ (1,3),\ (3,1),\ (4,4)$

Conclusion

For an equation of this type, it is most likely impossible to derive a general form of the solution, especially from previously proven facts, for almost any k, there is a finite number of solutions, but for specific k there can be a lot of them.

References

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